🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.NOS.4

Find Factors and Multiples

Lesson 6-7 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the greatest common factor and the least common multiple of two numbers, and use each to solve a real problem.
Language Objective: I can explain my choice using the words factor, multiple, common, greatest, and least.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Find Factors and Multiples | Factor | Multiple | Greatest common factor (GCF) | Least common multiple (LCM) | Factor pair | Prime factorization

👆 *Tap any word to see what it means and a picture.*

1. Factors __________ a number; multiples are what it __________.

2. A number that divides another exactly, with no remainder, is a __________.

3. What you land on counting by a number are its __________.

4. The largest number that divides both is their __________ __________ __________.

5 The smallest number both numbers reach is their __________ __________ __________.

6 Two factors that multiply to make the number are a __________ __________.

7 Writing a number as __________ numbers multiplied together is its prime factorization.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

What is the greatest number of combo packs the manager can make, and what goes in each one?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Find Factors and Multiples** — Listing what divides a number evenly (its factors) and what it divides into evenly (its multiples), then finding what two numbers share.
Hallar factores y múltiplos — Listar lo que divide un número exactamente (sus factores) y lo que el número forma al contar de esa cantidad (sus múltiplos), y luego hallar lo que dos números comparten.

- [ ] **Factor** — A whole number that divides another number evenly, with no remainder.
Factor — Un número entero que divide a otro exactamente, sin residuo.

- [ ] **Multiple** — The result of multiplying a number by 1, 2, 3, and so on — where you land when you skip-count by it.
Múltiplo — El resultado de multiplicar un número por 1, 2, 3, y así sucesivamente: donde caes al contar de esa cantidad.

- [ ] **Greatest common factor (GCF)** — The largest number that is a factor of two or more numbers. It answers "how many equal groups can I make?"
Máximo común divisor (MCD) — El número más grande que es factor de dos o más números. Responde "¿cuántos grupos iguales puedo formar?"

- [ ] **Least common multiple (LCM)** — The smallest number that is a multiple of two or more numbers. It answers "when will these line up again?"
Mínimo común múltiplo (mcm) — El número más pequeño que es múltiplo de dos o más números. Responde "¿cuándo volverán a coincidir?"

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The number of packs must be a ____ of both 48 and 36. Their common factors are 1, 2, 3, 4, 6, and ____. The greatest of these is the ____, so the manager makes ____ packs, each holding ____ pencils and 3 notebooks.
Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: The number of pods must divide BOTH 24 and 36 evenly.

Evidence: Students see the number of pods has to be a common factor of 24 and 36, so it must divide both with nothing left over, ruling out numbers that only divide one of them.

Reasoning: This evidence connects the definition of find factors and multiples to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The number of packs must be a ____ of both 48 and 36. Their common factors are 1, 2, 3, 4, 6, and ____. The greatest of these is the ____, so the manager makes ____ packs, each holding ____ pencils and 3 notebooks.**
- **My answer is ____.**

  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

  Mi afirmación es ____.
- **I know because ____.**

  Lo sé porque ____.
- **So ____.**

  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Find Factors and Multiples
2. **Notes 2:** Factor
3. **Notes 3:** Multiple
4. **Notes 4:** Greatest common factor (GCF)
5. **Notes 5:** Least common multiple (LCM)
6. **Notes 6:** Factor pair
7. **Notes 7:** Prime factorization
8. **Watch for this mistake:** *A common mistake is reaching for the wrong tool: listing FACTORS for a problem about repeating cycles. Ayasha's classes repeat every 6 and 8 weeks, so the weeks they land on are multiples — answering with the GCF of 2 claims a class that meets every 8 weeks happens in week 2. Use the size of the answer as a check: a GCF is never larger than the smaller number, and an LCM is never smaller than the larger one.*
9. **Try It 1:** *— When two numbers share no factor except 1, their GCF is 1 — which is a real answer, not a mistake.*
10. **Try It 2:** *— Factors are finite and sit at or below the number; multiples are infinite and sit at or above it. Knowing this checks your answer before you write it down.*